

# Flagged LLT polynomials and their applications

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Algebraic combinatorics and applications

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# Symmetric polynomials and functions

- Symmetric polynomials  $\mathbb{Q}[x_1, \dots, x_n]^{S_n}$
- Generators

$$e_r(x_1, \dots, x_n) = \sum_{1 \leq i_1 < i_2 < \dots < i_r \leq n} x_{i_1} \cdots x_{i_r}$$

$$e_1(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

$$e_2(x_1, x_2, x_3) = x_1x_2 + x_1x_3 + x_2x_3$$

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- Integer partitions of  $d$ .

# Partitions

## Definition

$n \in \mathbb{Z}_{>0}$ , a *partition of  $n$*  is  $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell > 0)$  such that  $\lambda_1 + \lambda_2 + \cdots + \lambda_\ell = n$ .

5  $\rightarrow$ 

|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|--|--|--|--|--|

$4 + 1 \rightarrow$  

$$3 + 2 \rightarrow \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$$

$3 + 1 + 1 \rightarrow$  

$$2 + 2 + 1 \rightarrow \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$$

$$2 + 1 + 1 + 1 \rightarrow \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}$$

$$1 + 1 + 1 + 1 + 1 \rightarrow$$

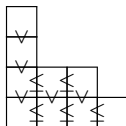
# Young Tableaux

## Definition

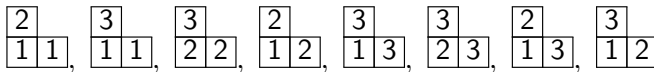
Filling of partition diagram of  $\lambda$  with numbers such that

- 1 strictly increasing up columns
- 2 weakly increasing along rows

Collection is called  $\text{SSYT}(\lambda)$ .



For  $\lambda = (2, 1)$ ,



# Schur polynomials

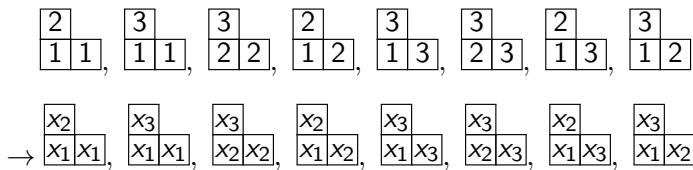
Associate a polynomial to  $\text{SSYT}(\lambda)$ .

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 |   | 3 |   | 3 |   | 2 |   | 3 |   | 3 |   | 2 |   | 3 |   |
| 1 | 1 | 1 | 1 | 2 | 2 | 1 | 2 | 1 | 3 | 2 | 3 | 1 | 3 | 1 | 2 |



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$$s_{(2,1)}(x_1, x_2, x_3) = x_1^2 x_2 + x_1^2 x_3 + x_2^2 x_3 + x_1 x_2^2 + x_1 x_3^2 + x_2 x_3^2 + 2x_1 x_2 x_3$$

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- $s_\lambda$  is a symmetric function.
- $\{s_\lambda\}_\lambda$  forms a basis for  $\Lambda$ .

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## Hidden Guide: Schur Positivity

“Naturally occurring” symmetric functions which are non-negative (coefficients in  $\mathbb{N}$ ) linear combinations in Schur polynomial basis are interesting since they could have representation-theoretic models.

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$$s_{\nu/\lambda} \cdot s_{\kappa/\mu} = \left( \sum_{T \in \text{SSYT}(\nu/\lambda)} x^T \right) \left( \sum_{S \in \text{SSYT}(\kappa/\mu)} x^S \right) = \sum_{\substack{T \in \text{SSYT}(\nu/\lambda) \\ S \in \text{SSYT}(\kappa/\mu)}} x^T x^S.$$

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- Littlewood-Richardson rule:  $s_\lambda s_\mu = \sum_\nu c_{\lambda\mu}^\nu s_\nu$  for  $c_{\lambda\mu}^\nu \in \mathbb{N}$ .
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$$s_{\begin{smallmatrix} \square & \square \\ & \square \end{smallmatrix}} = x_1^3 x_2 + 2x_1^2 x_2^2 + \dots = s_{\begin{smallmatrix} \square & \square & \square \\ & & \square \end{smallmatrix}} + s_{\begin{smallmatrix} \square & \square \\ & \square \end{smallmatrix}}$$

$$\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 1 \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 2 \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 2 & 2 \\ \hline 1 & 1 \\ \hline \end{array} \dots$$

- Straightforwardly,

$$s_{\nu/\lambda} \cdot s_{\kappa/\mu} = \left( \sum_{T \in \text{SSYT}(\nu/\lambda)} x^T \right) \left( \sum_{S \in \text{SSYT}(\kappa/\mu)} x^S \right) = \sum_{\substack{T \in \text{SSYT}(\nu/\lambda) \\ S \in \text{SSYT}(\kappa/\mu)}} x^T x^S.$$

$$s_{\begin{smallmatrix} \square & \square \\ & \square \end{smallmatrix}} \cdot s_{\begin{smallmatrix} \square & \square \\ & \square \end{smallmatrix}} = x_1^5 x_2^3 + \dots$$

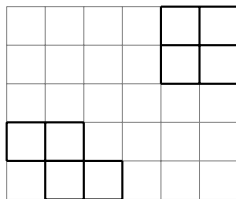
$$\left( \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 1 \\ \hline \end{array}, \begin{array}{|c|c|} \hline 2 & 2 \\ \hline 1 & 1 \\ \hline \end{array} \right) \dots$$

- $q$ -deformation?

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Let  $\nu = (\nu_{(1)}, \dots, \nu_{(k)})$  be a tuple of skew shapes. (Skew shape =  $\lambda \setminus \mu$ )

$$\nu = \left( \begin{array}{|c|c|c|} \hline \square & \square & \\ \hline \square & \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \right)$$



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|    |    |    |    |   |   |
|----|----|----|----|---|---|
| -4 | -3 | -2 | -1 | 0 | 1 |
| -3 | -2 | -1 | 0  | 1 | 2 |
| -2 | -1 | 0  | 1  | 2 | 3 |
| -1 | 0  | 1  | 2  | 3 | 4 |
| 0  | 1  | 2  | 3  | 4 | 5 |

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|       |       |       |  |       |       |
|-------|-------|-------|--|-------|-------|
|       |       |       |  | $b_3$ | $b_6$ |
|       |       |       |  | $b_5$ | $b_8$ |
|       |       |       |  |       |       |
| $b_1$ | $b_2$ |       |  |       |       |
|       | $b_4$ | $b_7$ |  |       |       |

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|       |       |       |  |       |       |
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Attacking pairs:  $(b_2, b_3), (b_3, b_4), (b_4, b_5), (b_4, b_6), (b_5, b_7), (b_6, b_7), (b_7, b_8)$

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|       |       |       |  |  |       |       |
| $b_1$ | $b_2$ |       |  |  |       |       |
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- A *semistandard tableau* on  $\nu$  is a map  $T: \nu \rightarrow \mathbb{Z}_+$  which restricts to a semistandard tableau on each  $\nu_{(i)}$ .

The *LLT polynomial* indexed by a tuple of skew shapes  $\nu$  is

$$\mathcal{G}_\nu(x; q) = \sum_{T \in \text{SSYT}(\nu)} x^T,$$

$$x^T = \prod_{a \in \nu} x_{T(a)}.$$

$T =$

|   |   |   |  |   |   |
|---|---|---|--|---|---|
|   |   |   |  | 5 | 6 |
|   |   |   |  | 1 | 1 |
|   |   |   |  |   |   |
| 2 | 4 |   |  |   |   |
|   | 3 | 5 |  |   |   |

$$x^T = x_1^2 x_2 x_3 x_4 x_5^2 x_6$$

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|---|---|---|--|---|---|
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non-inversion

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|   |   |   |  | 1 | 1 |  |
|   |   |   |  |   |   |  |
| 2 | 4 |   |  |   |   |  |
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where  $\text{inv}(T)$  is the number of attacking inversions in  $T$  and  $x^T = \prod_{a \in \nu} x_{T(a)}$ .

$T =$

|   |   |   |  |   |   |  |
|---|---|---|--|---|---|--|
|   |   |   |  | 5 | 6 |  |
|   |   |   |  | 1 | 1 |  |
|   |   |   |  |   |   |  |
| 2 | 4 |   |  |   |   |  |
|   | 3 | 5 |  |   |   |  |

inversion

$$x^T = x_1^2 x_2 x_3 x_4 x_5^2 x_6$$

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$$\text{inv}(T) = 4, \quad x^T = x_1^2 x_2 x_3 x_4 x_5^2 x_6$$



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- Thus,  $\mathcal{G}_\nu$  are useful to show combinatorial quantities are Schur-positive!

# Some notable occurrences of LLT Polynomials

- Haglund-Haiman-Loehr formula for Macdonald polynomials:

$$\tilde{H}_{\mu}(x; q, t) = t^{n(\mu)} \sum_R \left( \prod_{\substack{\square \\ u}} q^{a+1} t^l \right) \mathcal{G}_R(x; t^{-1}).$$

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- For  $G$  = incomparability graph for natural unit interval order (encoded by Dyck path  $P$ ), the  $t$ -chromatic symmetric polynomial is essentially a “signed” unicellular LLT polynomial (later), i.e.,

$$\chi_G(x; t) = (1 - t)^{-|V|} \mathcal{G}_{\nu(P)}[(1 - t)x; t].$$

# Flagged Tableaux

- Fix partition shape  $\lambda = (\lambda_1, \dots, \lambda_l)$  and *flag*  
 $\mathbf{b} = (b_1 \leq \dots \leq b_l) \in \mathbb{N}^l$ .

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- A *flagged semistandard Young tableau* of shape  $\lambda$  with flag  $\mathbf{b}$  is a  $T \in \text{SSYT}(\lambda)$  such that the entries of row  $i$  are bounded above by  $b_i$ .
- Denote the set of such tableaux via  $\text{FT}(\lambda, \mathbf{b})$ .
- E.g.,  $\lambda = (2, 1)$ ,  $\mathbf{b} = (1, 3)$ ,  $\text{FT}(\lambda, \mathbf{b}) =$

$$\begin{array}{|c|c|} \hline 2 & \\ \hline 1 & 1 \\ \hline \end{array} \leq 1 \quad \begin{array}{|c|c|} \hline 3 & \\ \hline 1 & 1 \\ \hline \end{array} \leq 1$$

# Flagged Schur functions (Lascoux-Schützenberger, Wachs)

- For partition  $\lambda$  and flag  $\mathbf{b}$ , the *flagged Schur function* is given by

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- $s_{\lambda, \mathbf{b}}(x)$  are examples of *Demazure characters* (Reiner-Shimozono, 1995).



## Demazure characters

The *Demazure operator*  $\pi_i$  acts on  $f \in \mathbb{K}[x_1^{\pm 1}, \dots, x_N^{\pm 1}]$  by

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The *Demazure characters* or *key polynomials* are constructed from

- $\mathcal{D}_\lambda = x^\lambda := x_1^{\lambda_1} \cdots x_N^{\lambda_N}$  for partition  $\lambda$ .
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$$\mathcal{D}_{520} = x_1^5 x_2^2$$

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- Demazure characters are characters of irreducible  $B$ -modules.

# Demazure atoms

*Demazure atoms* are defined the same as keys but with  $\hat{\pi}_i := \pi_i - 1$  in place of  $\pi_i$ :

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- Demazure atoms are related to Demazure characters by Bruhat order inclusion-exclusion:

$$\mathcal{D}_{520} = \mathcal{A}_{520}$$

$$\mathcal{D}_{250} = \mathcal{A}_{520} + \mathcal{A}_{250}$$

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- Demazure atoms form a dual basis of  $\mathbb{K}[x_1, \dots, x_N]$  to Demazure characters under a nice inner product.

# Recovering Symmetric Functions

- If  $\mathbf{b} = (n, \dots, n)$  for  $n = |\lambda|$ , then  $s_{\lambda, \mathbf{b}}(x) = s_{\lambda}(x_1, \dots, x_n)$ .



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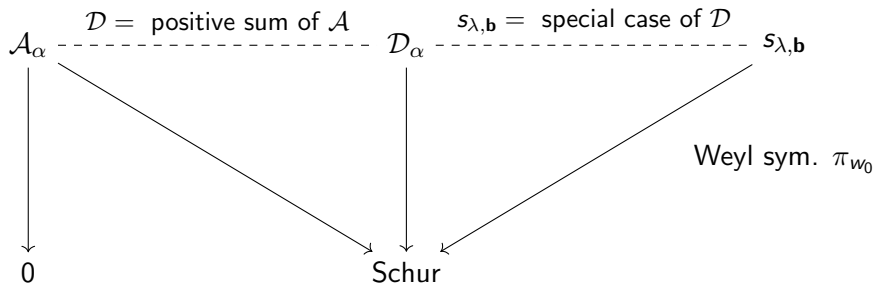
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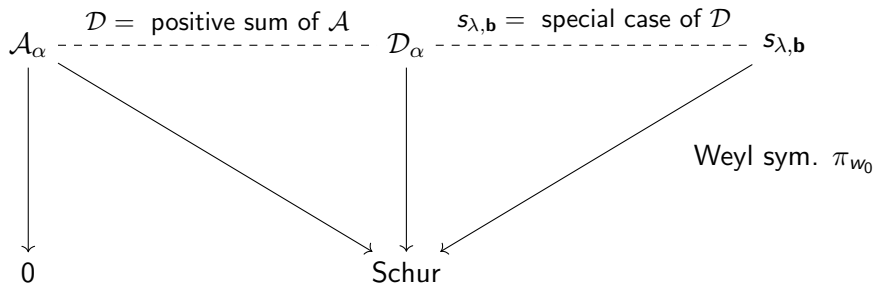
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$$\pi_{w_0}(\mathcal{A}_{\alpha}) = \begin{cases} s_{\alpha} & \alpha \text{ a partition,} \\ 0 & \text{else.} \end{cases}$$

# Summary of families



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## Question

Is there some “nice” family of polynomials that Weyl symmetrizes to LLT polynomials?

# Flagged LLT Polynomials (Blasiak-Haiman-Morse-Pun-S.)

- Let  $e_1, \dots, e_l$  be the row ends of  $\nu$ , ordered in reverse reading order.
- Fix flag  $\mathbf{b} = (b_1 \leq \dots \leq b_l)$ .
- $\text{FT}(\nu, \mathbf{b})$  = set of semistandard tableaux  $T$  on  $\nu$  satisfying  $T(e_i) \leq b_i$ .
- The *flagged LLT polynomial* indexed by  $\nu$  and  $\mathbf{b}$  is

$$\mathcal{G}_{\nu, \mathbf{b}}(x; t) = \sum_{T \in \text{FT}(\nu, \mathbf{b})} t^{\text{inv}(T)} x^T.$$

|  |       |       |  |  |  |       |
|--|-------|-------|--|--|--|-------|
|  |       |       |  |  |  | $e_3$ |
|  |       |       |  |  |  | $e_1$ |
|  |       |       |  |  |  |       |
|  | $e_4$ |       |  |  |  |       |
|  |       | $e_2$ |  |  |  |       |



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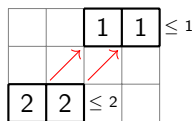
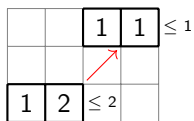
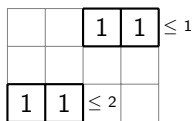
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|   |   |   |  |   |   |          |
|---|---|---|--|---|---|----------|
|   |   |   |  | 2 | 3 | $\leq 3$ |
|   |   |   |  | 1 | 1 | $\leq 1$ |
|   |   |   |  |   |   |          |
| 2 | 4 |   |  |   |   | $\leq 4$ |
|   | 1 | 2 |  |   |   | $\leq 2$ |

# Flagged LLT Polynomials

$$\nu = (\square\square, \square\square), \mathbf{b} = (1, 2)$$

T



$$\mathcal{G}_{r,\nu}(x; t) = x_1^4 + t x_1^3 x_2 + t^2 x_1^2 x_2^2$$

# Flagged LLT Polynomials

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- $\mathcal{G}_{\nu, \mathbf{b}}(x; t)$  Weyl symmetrizes to symmetric  $\mathcal{G}_{\nu}(x; t)$ .
- For  $\nu = (\lambda)$  a single shape,  $\mathcal{G}_{\nu, \mathbf{b}}(x; t) = s_{\lambda, \mathbf{b}}(x)$ .

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- $\mathcal{G}_{\nu, \mathbf{b}}(x; t)$  are conjecturally positive in terms of Demazure atoms.

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  - unbarred letters weakly increase in rows, strictly increase in columns.
  - barred letters strictly increase in rows, weakly increase in columns.
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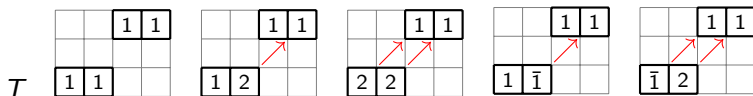
The *signed flagged LLT polynomial* indexed by  $\nu$  and flag  $\mathbf{b}$  is

$$\mathcal{G}_{\nu, \mathbf{b}}^\pm(x; t) = \sum_{T \in \text{FT}^\pm(\nu, \mathbf{b})} t^{\text{inv}(T)} (-t)^{-\#\text{bar}(T)} x^{|T|},$$

where  $|T|$  is the result of removing all bars from  $T$ .

# Signed flagged LLT polynomials

$$\nu = (\square\square, \square\square), \mathbf{b} = (1, 2)$$



$$\begin{aligned} \mathcal{G}_{\nu, \mathbf{b}}^{\pm}(x; t) &= x_1^4 + tx_1^3x_2 + t^2x_1^2x_2^2 - x_1^4 - tx_1^3x_2 \\ &= t^2x_1^2x_2^2 \end{aligned}$$

## Remark

Signed flagged LLTs are typically not positive in terms of Demazure atoms, nor are they generally even monomial positive.

# Flagged plethysm

Define *flagged plethysm*  $\Pi_{t,x}: \mathbb{k}[x_1, \dots, x_I] \rightarrow \mathbb{k}[x_1, \dots, x_I]$  to be the (over-determined) linear map

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**Theorem (Blasiak-Haiman-Morse-Pun-S., 2025+)**

$\Pi_{t,x}$  is well-defined.

$\Pi_{t,x}$  is a “nonsymmetric analogue” of the plethystic map  $f[X] \mapsto f[X/(1-t)]$  for symmetric function  $f$ .

# Applications

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- ② A “nonsymmetric shuffle theorem” expressed in terms of flagged LLT polynomials associated to flagged Dyck paths.
- ③ Tewari-Wilson-Zhang define chromatic nonsymmetric polynomials associated to  $d \times d$  Dyck paths starting with  $r$  north steps,  $\chi_{\mathbf{b}, \pi}$ . These are (essentially) examples of signed flagged LLT polynomials.



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- Macdonald provides a positivity conjecture about expanding  $J_\mu$  into a funny basis. What does it mean?

# Modified Macdonald polynomials

- Plethystically modified Macdonald polynomials

$$H_\mu(x; q, t) = J_\mu[x/(1-t); q, t].$$

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$$H_{31} = ts_4 + (1 + qt + q^2t)s_{31} + (q + tq^2)s_{22} + (q + q^2 + q^3t)s_{211} + q^3s_{1111}$$

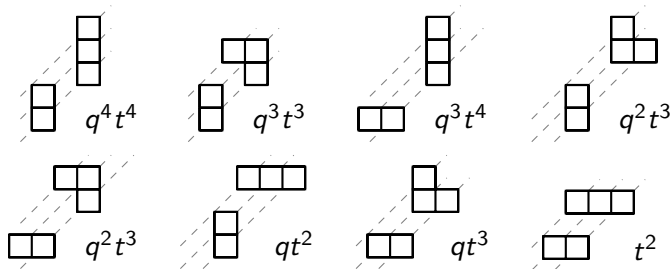
# Haglund-Haiman-Loehr formula example

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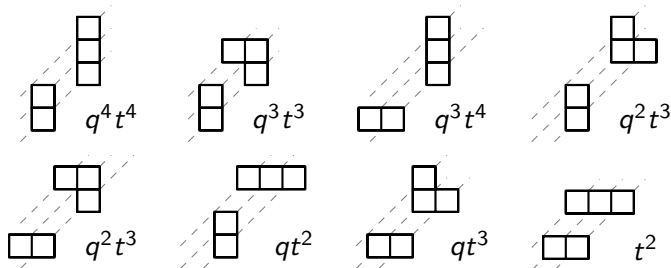
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For  $\mu = (3, 2)$ , we get 8 LLT terms:



Remark: Can permute the order of ribbons (but changes arm and leg statistics and LLTs.)

# Nonsymmetric Macdonald polynomials

The Cherednik operators  $Y_1, \dots, Y_N$  act on  $\mathbb{Q}(q, t)[x_1^{\pm 1}, \dots, x_N^{\pm 1}]$ .

$$T_i f = s_i f + (1 - t) x_i \frac{f - s_i f}{x_i - x_{i+1}}, \quad (\text{Demazure-Lusztig operators})$$

$$\Phi f = f(x_2, \dots, x_N, qx_1),$$

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- The  $\mathcal{E}_\alpha$ 's “Hecke symmetrize” to the integral form  $J_\mu$ 's.



## Nonsymmetric HHL formula example

$$\mathcal{E}_\alpha(x; q, t) = t^{n(\mu_+)} \sum_R \left( \prod_{\substack{u \\ \square}} q^{a+1} t^l \right) \mathcal{G}_{R, (1, 2, \dots, m)}^\pm(x; t^{-1}).$$

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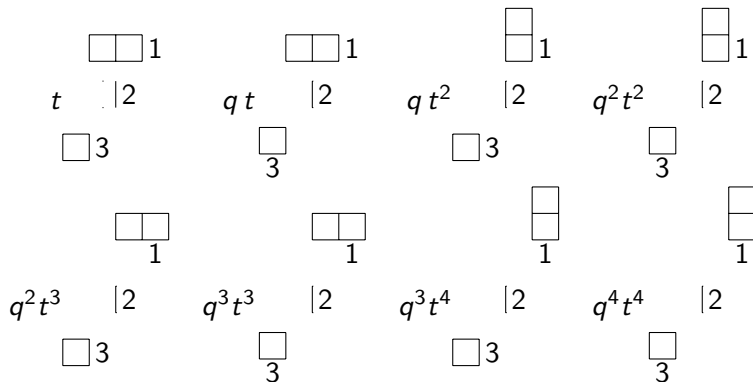
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For  $\alpha = (2, 0, 1)$ , we get 8 *signed* flagged LLT terms:



- Convention: bound boxes are always on the same content “diagonal.”
- Variation: bound boxes below shape bound shape strictly from below.

# The Missing Corner

$$\begin{array}{ccc} \mathcal{E}_\alpha(x_1, \dots, x_N; q, t) & \xrightarrow{\quad} & ? \\ \text{Hecke sym.} \downarrow & & \downarrow \text{Weyl sym.} \\ J_\mu(x; q, t) & \xrightarrow{\text{Plethysm}} & H_\mu(x; q, t) \end{array}$$

## Question

Is there a “nice” family of polynomials that Weyl symmetrizes to the modified Macdonald polynomials?

An important feature for “nice”? Positivity!

# Toward positivity

- Knop (2007) formulated a positivity conjecture for a stable version of  $\mathcal{E}_\alpha$  involving Kazhdan-Lusztig theory.
- Lapointe (2022) formulated another positivity conjecture for a stable version of  $\mathcal{E}_\alpha$ .
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- We can understand stabilizations in terms of signed flagged LLT combinatorics.
- Upshot: look at stabilizations for “positivity.”

# Filling in the missing corner

$$\mathcal{E}_\alpha(x_1, \dots, x_N; q, t) = \sum \text{signed flagged LLTs}$$

⋈ stabilize  
↓

$$\mathcal{J}_{\eta|\lambda}(x; q, t) = \sum \text{signed flagged LLTs}$$

Hecke sym.  
↓

$$J_{(\eta;\lambda)_+}(x; q, t) = \sum \text{signed LLTs}$$

→ ?

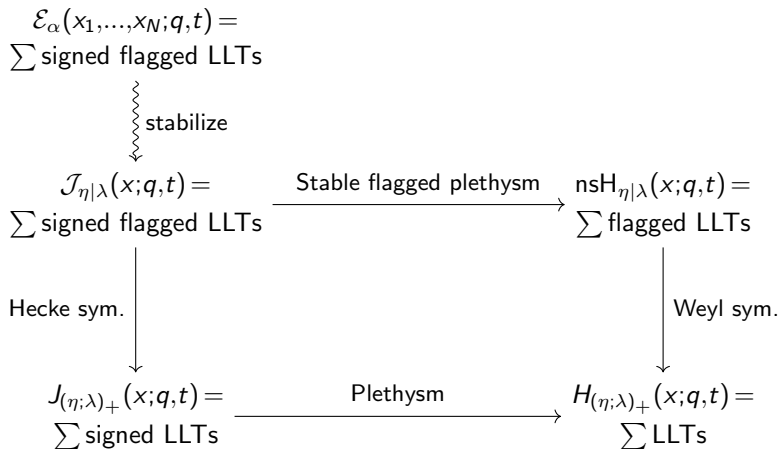
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Plethysm  
→



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# Modern Macdonald polynomials

Definition (Blasiak-Haiman-Morse-Pun-S., 2025+)

The *modern Macdonald polynomial* indexed by  $\eta \in \mathbb{N}^r$  and partition  $\lambda$  is

$$\text{nsH}_{\eta|\lambda}(x; q, t) = t^{n((\eta;\lambda)_+)} \sum_R \left( \prod_{\substack{u \\ \boxed{u}}} q^{a+1} t^l \right) \mathcal{G}_{R,(1,2,\dots,r)}(x; t^{-1})$$

For  $r = 2$ ,  $\eta = (2, 1)$ ,  $\lambda = \emptyset$ , the flagged LLT terms are

$$\begin{array}{cc}
 \begin{array}{c} \boxed{\phantom{0}} \boxed{\phantom{0}} \\ t \end{array} 1 & \begin{array}{c} \boxed{\phantom{0}} \\ \boxed{\phantom{0}} \\ q t^2 \end{array} 1 \\
 \begin{array}{c} \boxed{\phantom{0}} \\ 2 \end{array} & \begin{array}{c} \boxed{\phantom{0}} \\ 2 \end{array}
 \end{array}$$

$$\text{nsH}_{21|\emptyset}(x_1, x_2, x_3; q, t) = t x_1^3 + x_1^2 x_2 + q t x_1^2 x_2 + q t x_1 x_2^2 + q t x_1^2 x_3 + q x_1 x_2 x_3$$

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- $\text{nsH}_{\eta|\lambda}(x; q, t)$  Weyl symmetrizes to symmetric  $H_{\beta_+}(x; q, t)$ .
- We conjecture  $\text{nsH}_{\eta|\lambda}(x; q, t)$  is atom positive (would follow from flagged LLTs being atom positive).

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Thank you!

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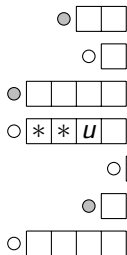
- $\mathcal{P}(r) = \mathbb{Q}(q, t)[x_1, \dots, x_r] \otimes \Lambda_{\mathbb{Q}(q, t)}(x_{r+1}, \dots)$ .
- Use a “stable” version of  $\mathcal{E}_\alpha$ , which we denote  $\mathcal{J}_{\eta|\lambda}$ .

$$\begin{array}{ccc}
 \mathcal{J}_{\eta|\lambda}(x; q, t) & \xrightarrow{\Pi_r} & \text{nsH}_{\eta|\lambda}(x; q, t) \\
 \downarrow \text{Hecke sym.} & & \downarrow \text{Weyl sym.} \\
 J_{(\eta; \lambda)_+}(x; q, t) & \xrightarrow{\text{Plethysm}} & H_{(\eta; \lambda)_+}(x; q, t)
 \end{array}$$

- $\mathcal{J}_{\eta|\lambda}(x; q, t)$  and  $\text{nsH}_{\eta|\lambda}(x; q, t)$  for  $(\eta|\lambda) \in \mathbb{N}^r \times \text{Par}$  form bases of  $\mathcal{P}(r)$ .
- $(\eta; \lambda)_+ =$  partition rearrangement of concatenating  $(\eta; \lambda)$ .
- $\Pi_r$  is “ $r$ -nonsymmetric plethysm”

$$\Pi_r(f(x_1, \dots, x_r) \otimes g(x)) = (\Pi_{t,x} f(x_1, \dots, x_r)) \otimes g[x/(1-t)] .$$

# Arms and legs



$$a(u) = 2, l(u) = 3$$